

1. Administrative & Study Identification

Full Study Title: A Study of Combination Therapies With or Without Pembrolizumab (MK-3475) and/or Chemotherapy in Participants With Advanced Esophageal Cancer (MK-3475-06A) **Short Title/Acronym:** KEYMAKER-U06 Substudy 06A **Protocol Number:** MK-3475-06A **Master Protocol:** MK-3475-U06 **NCT Number:** NCT05342636 **Posting ID:** 551129514646302796 **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Sponsor Name:** Merck Sharp & Dohme LLC (Merck) **Investigational Product:** Pembrolizumab (MK-3475) in combination with investigational agents (such as MK-4280A, MK-4830, lenvatinib, sacituzumab tirumotecan) and/or chemotherapy (paclitaxel or irinotecan [Camptosar, Onivyde]) **Phase of Development:** Phase 1 / Phase 2 (PHASE1, PHASE2) **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate safety and tolerability during the safety lead-in phase (assessed by Dose-Limiting Toxicities [DLTs], Adverse Events [AEs], and study treatment discontinuation due to AEs) and to determine the Objective Response Rate (ORR) in the efficacy phase. * **Secondary Objectives:** To evaluate Progression-Free Survival (PFS), Duration of Response (DOR), Overall Survival (OS), and the ongoing safety/tolerability of the combination therapies.

2.2 Background & Rationale To address the critical unmet medical need in patients with advanced esophageal cancer that progresses after first-line chemotherapy. Despite advances, these patients continue to have limited treatment options and a poor prognosis. This umbrella platform study aims to evaluate novel immunotherapy and chemotherapy combinations in patients naïve to programmed cell death 1 protein (PD-1)/programmed cell death ligand 1 (PD-L1) based treatments to improve clinical outcomes.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional (Umbrella Platform Design) * **Control Type:** Active Comparator / Experimental Arms * **Blinding:** Open-label * **Randomization:** Parallel group assignment

3.2 Planned Enrollment & Duration * **Target N\$:** Approximately 120 patients * **Number of Sites:** Multicenter (Global) * **Estimated Study Start:** July 2022 * **Trial Status:** Active but not currently

recruiting * **Estimated Primary Completion:** Ongoing/Completed tracking depending on specific substudy arm closure

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically or cytologically confirmed diagnosis of metastatic or locally advanced unresectable Esophageal Squamous Cell Carcinoma (ESCC). 2. Experienced investigator-documented radiographic or clinical disease progression on one prior line of standard therapy and naïve to anti-PD-1/L1 therapy. 3. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. Has an evaluable baseline tumor sample (newly obtained or archival) for analysis. 5. Has adequately controlled blood pressure (BP) with or without antihypertensive medications. 6. Participants who have adverse events (AEs) due to previous anticancer therapies must have recovered to ≤Grade 1 or baseline. (Participants with endocrine-related AEs adequately treated with hormone replacement or participants with ≤Grade 2 neuropathy are eligible).

4.2 Exclusion Criteria 1. Direct invasion into adjacent organs such as the aorta or trachea. 2. Has experienced weight loss >10% over approximately 2 months prior to the first dose of study therapy. 3. Has received an investigational agent or has used an investigational device within 4 weeks prior to study intervention administration. 4. Diagnosis of immunodeficiency or is receiving chronic systemic steroid therapy or any other form of immunosuppressive therapy within 7 days prior to the first dose of study medication. 5. Known additional malignancy that is progressing or has required active treatment within the past 3 years (except basal cell/squamous cell carcinoma of the skin or carcinoma in situ that has undergone potentially curative therapy). 6. Known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 7. Active autoimmune disease requiring systemic treatment in the past 2 years. 8. History of human immunodeficiency virus (HIV), Hepatitis B, or known active Hepatitis C virus infection. 9. History of allogenic tissue/solid organ transplant. 10. Clinically significant cardiovascular disease within 12 months from the first dose of study intervention. 11. Known gastrointestinal (GI) malabsorption or any other condition that may affect the absorption of lenvatinib. 12. Risk for significant GI bleeding, such as a serious nonhealing wound, peptic ulcer, or bone fracture within 28 days prior to allocation; or significant bleeding disorders, vasculitis, or a significant GI bleeding episode within 12 weeks prior to allocation.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Multiple arms administering Pembrolizumab combined with investigator's choice

of chemotherapy (irinotecan [Camptosar, Onivyde] or paclitaxel), MK-4280A, MK-4830 + chemotherapy, or MK-4830 + lenvatinib. * **Note:** With protocol amendment 5 (effective: 17-November-2023), enrollment in study arms "Pembrolizumab plus MK-4830 plus Chemotherapy" and "Pembrolizumab plus MK-4830 plus lenvatinib" is discontinued. * **Route of Administration:** Intravenous (for Pembrolizumab, chemotherapies, and IV investigational agents) and Oral (for lenvatinib). * **Duration of Treatment:** Continues until documented disease progression, unacceptable toxicity, investigator decision, or participant withdrawal.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard supportive care medications (e.g., anti-emetics, standard antihypertensives for controlled BP). * **Prohibited:** Concurrent systemic immunosuppressive therapy (e.g., chronic systemic steroids) and other investigational therapies.

6. Study Timeline & Visit Schedule

- **Screening** (Days -28 to 0): Informed Consent, Medical History, Tumor sample acquisition, Baseline imaging
 - **Baseline** (Day 0): Allocation/Randomization, First Dose
 - **Interim** (Cycle-based): Safety Labs, PK/PD evaluations, Efficacy Assessment (Tumor imaging per RECIST 1.1)
 - **End of Study:** Discontinuation visit, Safety follow-up, Survival follow-up
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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **Safety Lead-in Phase:** 1. Number of Participants Experiencing a Dose-limiting Toxicity (DLT) (defined as any drug-related AE per NCI CTCAE v5.0 resulting in dose change/delay). 2. Number of Participants Experiencing an Adverse Event (AE). 3. Number of Participants Who Discontinue Study Treatment Due to an AE. * **Efficacy Phase:** Objective Response Rate (ORR). * **Measurement Method:** ORR assessed by Blinded Independent Central Review (BICR) using RECIST v1.1. DLTs/AEs graded per NCI CTCAE v5.0.

7.2 Secondary & Exploratory Endpoints * Progression-Free Survival (PFS) per RECIST v1.1 by BICR * Duration of Response (DOR) per RECIST v1.1 by BICR * Overall Survival (OS)

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous evaluation of untoward medical occurrences temporally associated with the study intervention, graded per NCI CTCAE v5.0.

- **Serious Adverse Events (SAEs):** Reported continuously for events resulting in death, life-threatening conditions, or requiring inpatient hospitalization.
 - **Data Monitoring Committee (DMC):** Evaluates safety lead-in data (observing if ≥ 4 patients experience DLTs out of the initial 10 per arm) to determine efficacy phase progression.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** All allocated participants (Intent-to-Treat for efficacy); All participants who received at least one dose of study medication (Safety Set).
 - **Sample Size Justification:** Up to 30 patients per arm (inclusive of 10 safety lead-in patients) based on DLT incidence thresholds.
 - **Handling of Missing Data:** Standard oncology censoring rules applied for time-to-event endpoints (PFS, OS, DOR).
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Routine remote and onsite monitoring visits focusing on DLT observation periods and RECIST assessments.
 - **Audit Trail:** Secure eCRF entry, tumor imaging archiving for BICR, and comprehensive data clarification mechanisms.
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11. Study Identifiers & Governance

- **Primary ID:** {{MK-3475-06A}}
 - **Master Protocol:** {{MK-3475-U06}}
 - **Registry Number:** {{NCT05342636}}
 - **Posting ID:** {{551129514646302796}}
 - **Sponsor/Lead Organization:** {{Merck Sharp & Dohme LLC (Merck)}}
 - **Study Title:** {{KEYMAKER-U06 Substudy 06A}}
 - **Official Regulatory Title:** {{A Phase 1/2 Open-Label, Umbrella Platform Design Study of Investigational Agents With or Without Pembrolizumab (MK-3475) and/or Chemotherapy in Participants With Advanced Esophageal Cancer naïve to PD-1/PD-L1 Treatment (KEYMAKER-U06): Substudy 06A}}
 - **Trial Status:** {{ACTIVE_NOT_RECRUITING}}
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12. Clinical Framework (Oncology Specific)

- **Primary Condition:** {{Advanced Esophageal Cancer (Esophageal Squamous Cell Carcinoma)}}
- **Diagnosis Threshold:** {{Histologically or cytologically confirmed metastatic or locally advanced unresectable ESCC}}
- **Medical Methodology:** {{Anti-PD-1 immunotherapy combined with targeted therapy/chemotherapy}}
- **Optimization Target:** {{Tumor response, Progression-free survival, and safety/tolerability}}

13. Study Procedures & Methodology

- **Management Pathway:** {{Systemic antineoplastic therapy per assigned platform arm}}
- **Education Protocol (HCP):** {{Investigator training on DLT criteria and NCI CTCAE v5.0 grading}}
- **Education Protocol (Patient):** {{Informed consent process, AE self-reporting instructions}}
- **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for efficacy endpoints}}

14. Observation & Follow-Up Schedule

- **Total Duration:** {{Follow-up until confirmed PD, death, or trial end}}
- **Onsite Visit Cadence:** {{Matched to treatment administration cycles (e.g., Q3W) and imaging schedules}}
- **Remote Monitoring Frequency:** {{As required for survival follow-up phases}}
- **Checklist Review Interval:** {{Every treatment cycle for safety and specified intervals for RECIST 1.1 imaging}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				